

E030

BSCNE-AWS Capstone Project

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

In a capstone VPC, you launch a web instance with a public IP address in a subnet whose route table has only the local VPC route. The instance type is t3.micro and it runs Amazon Linux. Browsers cannot reach it. What is missing?

- A. A larger instance type.
- B. Nothing; a public IP is enough.
- C. A 0.0.0.0/0 route to an attached internet gateway.
- D. A NAT gateway route.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A database instance must download OS updates from the internet but must never accept connections from the internet. What design fits?

- A. Private subnet with no internet path.
- B. Public subnet with an internet gateway route.
- C. Public subnet, protected only by a strong password.
- D. Private subnet with a default route to a NAT gateway in a public subnet.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A task asks you to block all traffic from 198.51.100.0/24 to a web subnet. A teammate opens the instances' security group, which has 12 rules and was created by the default wizard. What is the problem with that approach?

- A. Security groups are allow-only; use a network ACL deny rule on the subnet.
- B. Delete the security group.
- C. The group has too many rules.
- D. Add a deny rule with a lower number in the security group.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A network ACL has rule 90: deny all from 198.51.100.0/24, and rule 100: allow TCP 443 from 0.0.0.0/0. A client at 198.51.100.7 sends HTTPS. What happens?

- A.** Denied by rule 90.
- B.** Allowed by rule 100.
- C.** Allowed, because NACLs are stateful.
- D.** Allowed, since allow rules beat deny rules.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A user's group policy allows s3:GetObject on the reports bucket. A second policy attached to the user explicitly denies all S3 actions on that bucket. The user is in the finance department and joined in March. Can the user read objects?

- A.** Yes, because finance needs reports.
- B.** No; the explicit deny wins.
- C.** Yes; the more specific action wins.
- D.** Yes; the group allow wins.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A user has policies granting only S3 and EC2 actions. The user tries to read a DynamoDB table. What is the result and why?

- A.** Allowed, since nothing denies it.
- B.** Denied by the implicit default deny.
- C.** Allowed, because EC2 permissions cover all services.
- D.** Denied by an explicit deny.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An application on EC2 must write to an S3 bucket. The quickest idea is to paste an IAM user's access key into the app's config file. The app is written in Python and runs on port 8080. What is the correct approach?

- A. Change the app's port first.
- B. Paste the key but restrict file permissions.
- C. Attach an IAM role with scoped S3 write permission to the instance.
- D. Use the root account keys for reliability.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

You must update a CloudFormation stack's database settings, and some changes might replace the database. The stack has 40 resources and a descriptive name. What should you do before executing the update?

- A. Delete and recreate the stack.
- B. Rename the stack first.
- C. Edit the database directly in the console.
- D. Create and review a change set to see which resources will be modified or replaced.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A partner company needs to read one folder of reports in an S3 bucket. The bucket has Block Public Access turned on and was created by a colleague who has since changed teams. What is the best way to share?

- A. Ask the original creator to decide.
- B. Keep Block Public Access on and grant the partner's account scoped read access with a bucket policy (or presigned URLs).
- C. Turn off Block Public Access and make the bucket public.
- D. Grant the partner full S3 access in your account.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

You create a /26 subnet in a VPC for application servers. The VPC is in us-east-1 and has three availability zones. How many usable IP addresses does the subnet provide?

- A. 62
- B. 64
- C. 59
- D. 26

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

You need a VPC subnet for 100 instances. What is the smallest prefix that works?

- A. /25
- B. /26
- C. /28
- D. /24

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A security group was changed yesterday to allow SSH from anywhere. The team's dashboard shows CPU graphs in CloudWatch, and the account has a monthly budget alert. Which service shows who made the change?

- A. AWS Budgets.
- B. Amazon CloudWatch metrics.
- C. AWS Config.
- D. AWS CloudTrail.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

In a capstone VPC, you launch a web instance with a public IP address in a subnet whose route table has only the local VPC route. ~~The instance type is t3.micro and it runs Amazon Linux.~~ Browsers cannot reach it. What is missing?

Target word: why a public-IP instance is unreachable **Decides:** No route to an internet gateway.

A	Camouflage	Size does not affect routing.
B	Dropped element	Drops the route requirement.
C	Correct	Without it the subnet is not public.
D	Confusion set	NAT gives outbound-only access for private subnets.

2. Answer D Concept 1 • transfer

A database instance must download OS updates from the internet but must never accept connections from the internet. What design fits?

Target word: reverse: outbound-only access **Decides:** Private subnet with outbound path.

A	Dropped element	Cannot download updates.
B	Inverted operation	Makes it reachable if it has a public IP.
C	Adjacent category	Exposes the database.
D	Correct	Allows outbound, blocks unsolicited inbound.

3. Answer A Concept 2

A task asks you to block all traffic from 198.51.100.0/24 to a web subnet. A teammate opens the instances' security group, ~~which has 12 rules and was created by the default wizard.~~ What is the problem with that approach?

Target word: where to place a deny **Decides:** Security groups cannot deny.

A	Correct	NACLs support explicit deny.
B	Inverted operation	Removes protection instead of blocking.
C	Camouflage	Rule count is not the issue.
D	Confusion set	Numbered rules and denies belong to NACLs.

4. Answer A Concept 2 • transfer

A network ACL has rule 90: deny all from 198.51.100.0/24, and rule 100: allow TCP 443 from 0.0.0.0/0. A client at 198.51.100.7 sends HTTPS. What happens?

Target word: NACL order **Decides:** Lowest number evaluated first.

A	Correct	Rule 90 matches before rule 100.
B	Inverted operation	Higher numbers are evaluated later.
C	Confusion set	NACLs are stateless; state is not the issue here.
D	Absolute language	Order decides in NACLs.

5. Answer B Concept 3

A user's group policy allows s3:GetObject on the reports bucket. A second policy attached to the user explicitly denies all S3 actions on that bucket. ~~The user is in the finance department and joined in March.~~ Can the user read objects?

Target word: result of conflicting policies **Decides:** Explicit deny overrides allow.

A	Camouflage	Department does not decide.
B	Correct	Explicit deny always overrides.
C	Adjacent category	Specificity does not override a deny.
D	Inverted operation	Allows never beat explicit denies.

6. Answer B Concept 3 · transfer

A user has policies granting only S3 and EC2 actions. The user tries to read a DynamoDB table. What is the result and why?

Target word: no statement at all **Decides:** Default is deny.

A	Inverted operation	Absence of deny is not an allow.
B	Correct	No allow exists for DynamoDB.
C	Adjacent category	Permissions are per service and action.
D	Confusion set	There is no explicit deny; it is implicit.

7. Answer C Concept 4

An application on EC2 must write to an S3 bucket. The quickest idea is to paste an IAM user's access key into the app's config file. ~~The app is written in Python and runs on port 8080.~~ What is the correct approach?

Target word: credential method on EC2 **Decides:** Long-lived keys on a server are a risk.

A	Camouflage	Port does not decide credential handling.
B	Dropped element	Still a long-lived key on the server.
C	Correct	Roles supply temporary credentials automatically.
D	Inverted operation	The riskiest credential possible.

8. Answer D Concept 5

You must update a CloudFormation stack's database settings, and some changes might replace the database. ~~The stack has 40 resources and a descriptive name.~~ What should you do before executing the update?

Target word: safe stack update **Decides:** Preview whether resources will be replaced.

A	Adjacent category	Destroys resources unnecessarily.
B	Camouflage	Names do not affect impact.
C	Inverted operation	Causes drift.
D	Correct	Change sets preview impact.

9. Answer B Concept 6

A partner company needs to read one folder of reports in an S3 bucket. The bucket has Block Public Access turned on ~~and was created by a colleague who has since changed teams~~. What is the best way to share?

Target word: sharing without public exposure **Decides:** Grant scoped access to a specific party.

A	Camouflage	Ownership history does not decide the method.
B	Correct	Specific access, no public exposure.
C	Inverted operation	Exposes everything to everyone.
D	Adjacent category	Far broader than one folder.

10. Answer C Concept 7

You create a /26 subnet in a VPC for application servers. ~~The VPC is in us-east-1 and has three availability zones~~. How many usable IP addresses does the subnet provide?

Target word: usable addresses in an AWS subnet **Decides:** 64 – 5 reserved.

A	Adjacent category	Traditional minus-2 rule, not AWS.
B	Dropped element	Leaves out reserved addresses.
C	Correct	$2^6 = 64$; AWS reserves 5.
D	Wrong stem number	The prefix length.

11. Answer A Concept 7 · transfer

You need a VPC subnet for 100 instances. What is the smallest prefix that works?

Target word: solve for prefix **Decides:** /25 gives 123 usable; /26 gives 59.

A	Correct	$128 - 5 = 123 \geq 100$.
B	Dropped element	Only 59 usable.
C	Inverted operation	Larger prefix means fewer addresses: 11 usable.
D	Adjacent category	Fits, but not the smallest.

12. Answer D Concept 8

A security group was changed yesterday to allow SSH from anywhere. ~~The team's dashboard shows CPU graphs in CloudWatch, and the account has a monthly budget alert~~. Which service shows who made the change?

Target word: who made an API change **Decides:** API activity by identity.

A	Confusion set	Tracks cost, not changes.
B	Camouflage	The dashboard is in the stem, but metrics do not record who changed what.
C	Adjacent category	Shows configuration history and compliance; CloudTrail names the caller.
D	Correct	Records API calls and the identity that made them.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Public vs. private subnets	A public subnet has a route to an internet gateway (and instances need public IPs); a private subnet has no such route and reaches the internet outbound-only through a NAT gateway.	Thinking a public IP alone makes an instance reachable.	An instance that cannot reach, or should not be reachable from, the internet.
2	Security groups vs. network ACLs	Security groups are stateful, instance-level, and allow-only; network ACLs are stateless, subnet-level, support allow and deny, and are evaluated by lowest rule number first.	Trying to write a deny rule in a security group.	A need to block a specific address, or an ordered numbered rule list.
3	IAM policy evaluation	Everything is denied by default; an explicit allow grants access; an explicit deny anywhere overrides any allow.	Believing an allow in one policy beats a deny in another.	Two policies that conflict for the same action.
4	Roles instead of access keys	Applications on EC2 should use an IAM role (instance profile) that supplies short-lived credentials, not long-lived access keys stored on the server or in code.	Copying a user's access keys onto an instance.	An app on EC2 that needs to call AWS services.
5	CloudFormation stacks, change sets, and drift	CloudFormation templates declare resources; change sets preview what an update will do; drift detection shows resources changed outside the stack.	Editing stack resources by hand in the console.	A stack update that could replace resources, or a console change.
6	S3 data protection	Keep S3 Block Public Access on unless public access is truly required; grant access with scoped bucket policies or IAM, and enable default encryption.	Making a bucket public to share one file.	A request to share S3 data with a specific party.
7	AWS subnet addressing	AWS reserves 5 addresses in every subnet, so usable addresses = $2^{(32 - \text{prefix})} - 5$.	Using the traditional "minus 2" rule inside a VPC.	A VPC subnet prefix and an instance count.
8	CloudTrail vs. CloudWatch vs. Config	CloudTrail records who made which API call; CloudWatch collects metrics, logs, and alarms; AWS Config records resource configuration history and evaluates compliance rules.	Looking for "who deleted it" in CloudWatch metrics.	Who / how much / is it compliant.

Course scope (public description): In the BSCNE-AWS Capstone Project, students choose from a fixed set of tasks that they might encounter in the workplace. Examples might include configuring network components, applying security policies to a set of cloud resources, or automating a network deployment using orchestration tools. Students will submit their tasks for evaluation in a working, deployed environment where an evaluator can

check their results by inspecting configurations or logging into machines or services to test functionality. All tasks will be completed in an AWS environment using either the AWS CLI or web-based tools and standard AWS platform components.